NONUNIQUE GENUS-ZERO MINIMAL MODELS HAVE PROJECTIVE-LINE SPECIAL FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

Allowed standard prerequisites (editorial).

- DVR models of smooth projective curves, common resolutions of regular surfaces, contraction of exceptional curves of the first kind, strict transforms and regular local rings. Original numerical-type and minimality support retained.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

Source reference. Original lemma 0C7C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15327–15327. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C8U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15379–15379. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C8T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15378–15378. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C5X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15277–15277. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original definition 06LG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8465–8465. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 0C2I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15158–15158. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C5Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15279–15279. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0CA3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15423–15423. Stable Stacks locator for declared

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Source reference. Original equation 0C65: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15285–15285. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 0C64: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15284–15284. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C66: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15286–15286. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C6P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15304–15304. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original proposition 0C6U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15309–15309. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C5S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15272–15272. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0BFP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14392–14392. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C5R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15271–15271. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C2N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15163–15163. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C5J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15264–15264. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 868–868. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0B8Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14156–14156. Stable Stacks locator for declared

standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 080E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10179–10179. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0807: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10172–10172. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

1. Numerical types

Part of the arguments will involve the combinatorics of the following data structures.

Definition 1.1. A *numerical type* T is given by

$$n, m_i, a_{ij}, w_i, g_i$$

where $n \geq 1$ is an integer and m_i, a_{ij}, w_i, g_i are integers for $1 \leq i, j \leq n$ subject to the following conditions

(1) $m_i > 0$, $w_i > 0$, $g_i \geq 0$,
(2) the matrix $A = (a_{ij})$ is symmetric and $a_{ij} \geq 0$ for $i \neq j$,
(3) there is no proper nonempty subset $I \subset \{1, \ldots, n\}$ such that $a_{ij} = 0$ for $i \in I$, $j \notin I$,
(4) for each i we have $\sum_j a_{ij} m_j = 0$, and
(5) $w_i | a_{ij}$.

This is obviously a somewhat annoying type of structure to work with, but it is exactly what shows up in special fibres of proper regular models of smooth geometrically connected curves. Of course we only care about these types up to reordering the indices.

Definition 1.2. We say two numerical types n, m_i, a_{ij}, w_i, g_i and $n', m'_i, a'_{ij}, w'_i, g'_i$ are *equivalent types* if there exists a permutation σ of $\{1, \ldots, n\}$ such that $m_i = m'_{\sigma(i)}$, $a_{ij} = a'_{\sigma(i)\sigma(j)}$, $w_i = w'_{\sigma(i)}$, and $g_i = g'_{\sigma(i)}$.

A numerical type has a genus.

Lemma 1.3. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type. Then the expression*

$$g = 1 + \sum m_i (w_i(g_i - 1) - \frac{1}{2} a_{ii})$$

is an integer.

Proof. To prove g is an integer we have to show that $\sum a_{ii} m_i$ is even. This we can see by computing modulo 2 as follows

$$\begin{aligned}
\sum\nolimits_i a_{ii} m_i &\equiv \sum\nolimits_{i,\ m_i \text{ odd}} a_{ii} m_i \\
&\equiv \sum\nolimits_{i,\ m_i \text{ odd}} \sum\nolimits_{j \neq i} a_{ij} m_j \\
&\equiv \sum\nolimits_{i,\ m_i \text{ odd}} \sum\nolimits_{j \neq i,\ m_j \text{ odd}} a_{ij} m_j \\
&\equiv \sum\nolimits_{i<j,\ m_i \text{ and } m_j \text{ odd}} a_{ij}(m_i + m_j) \\
&\equiv 0
\end{aligned}$$

where we have used that $a_{ij} = a_{ji}$ and that $\sum_j a_{ij} m_j = 0$ for all i. □

Definition 1.4. We say n, m_i, a_{ij}, w_i, g_i is a *numerical type of genus* g if $g = 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ is the integer from Lemma 1.3.

We will prove below (Lemma 0C7C) that the genus is almost always ≥ 0. But you can have numerical types with negative genus.

Lemma 1.5. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type of genus g. If $n = 1$, then $a_{11} = 0$ and $g = 1 + m_1w_1(g_1 - 1)$. Moreover, we can classify all such numerical types as follows*

1. *If $g < 0$, then $g_1 = 0$ and there are finitely many possible numerical types of genus g with $n = 1$ corresponding to factorizations $m_1w_1 = 1 - g$.*
2. *If $g = 0$, then $m_1 = 1$, $w_1 = 1$, $g_1 = 0$ as in Lemma 1.14.*
3. *If $g = 1$, then we conclude $g_1 = 1$ but m_1, w_1 can be arbitrary positive integers; this is case (0C8U) of Lemma 0C8T.*
4. *If $g > 1$, then $g_1 > 1$ and there are finitely many possible numerical types of genus g with $n = 1$ corresponding to factorizations $m_1w_1(g_1 - 1) = g - 1$.*

Proof. The lemma proves itself. □

Lemma 1.6. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type of genus g. If $n > 1$, then $a_{ii} < 0$ for all i.*

Proof. Lemma 0C5X applies to the matrix A. □

Lemma 1.7. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type of genus g. Assume $n > 1$. If i is such that the contribution $m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ to the genus g is < 0, then $g_i = 0$ and $a_{ii} = -w_i$.*

Proof. Follows immediately from Lemma 1.6 and $w_i > 0$, $g_i \geq 0$, and $w_i | a_{ii}$. □

Definition 1.8. Let n, m_i, a_{ij}, w_i, g_i be a numerical type. We say i is a (-1)-*index* if $g_i = 0$ and $a_{ii} = -w_i$.

We can "contract" (-1)-indices.

Lemma 1.9. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type T. Assume n is a (-1)-index. Then there is a numerical type T' given by $n', m'_i, a'_{ij}, w'_i, g'_i$ with*

1. *$n' = n - 1$,*
2. *$m'_i = m_i$,*
3. *$a'_{ij} = a_{ij} - a_{in}a_{jn}/a_{nn}$,*
4. *$w'_i = w_i/2$ if a_{in}/w_n even and a_{in}/w_i odd and $w'_i = w_i$ else,*
5. *$g'_i = \frac{w_i}{w'_i}(g_i - 1) + 1 + \frac{a_{in}^2 - w_n a_{in}}{2w'_i w_n}$.*

Moreover, we have $g = g'$.

Proof. Observe that $n > 1$ for example by Lemma 1.5 and hence $n' \geq 1$. We check conditions (1) – (5) of Definition 1.1 for $n', m'_i, a'_{ij}, w'_i, g'_i$.

Condition (1) is immediate.

Condition (2). Symmetry of $A' = (a'_{ij})$ is immediate and since $a_{nn} < 0$ by Lemma 1.6 we see that $a'_{ij} \geq a_{ij} \geq 0$ if $i \neq j$.

Condition (3). Suppose that $I \subset \{1, \ldots, n-1\}$ such that $a'_{ii'} = 0$ for $i \in I$ and $i' \in \{1, \ldots, n-1\} \backslash I$. Then we see that for each $i \in I$ and $i' \in I'$ we have $a_{in}a_{i'n} = 0$. Thus either $a_{in} = 0$ for all $i \in I$ and $I \subset \{1, \ldots, n\}$ is a contradiction for property (3) for T, or $a_{i'n} = 0$ for all $i' \in \{1, \ldots, n-1\} \setminus I$ and $I \cup \{n\} \subset \{1, \ldots, n\}$ is a contradiction for property (3) of T. Hence (3) holds for T'.

Condition (4). We compute

$$\sum_{j=1}^{n-1} a'_{ij}m_j = \sum_{j=1}^{n-1}(a_{ij}m_j - \frac{a_{in}a_{jn}m_j}{a_{nn}}) = -a_{in}m_n - \frac{a_{in}}{a_{nn}}(-a_{nn}m_n) = 0$$

as desired.

Condition (5). We have to show that w'_i divides $a_{in}a_{jn}/a_{nn}$. This is clear because $a_{nn} = -w_n$ and $w_n|a_{jn}$ and $w_i|a_{in}$.

To show that $g = g'$ we first write

$$\begin{aligned} g &= 1 + \sum_{i=1}^{n} m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) \\ &= 1 + \sum_{i=1}^{n-1} m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) - \frac{1}{2}m_n w_n \\ &= 1 + \sum_{i=1}^{n-1} m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii} - \frac{1}{2}a_{in}) \end{aligned}$$

Comparing with the expression for g' we see that it suffices if

$$w'_i(g'_i - 1) - \frac{1}{2}a'_{ii} = w_i(g_i - 1) - \frac{1}{2}a_{in} - \frac{1}{2}a_{ii}$$

for $i \leq n-1$. In other words, we have

$$g'_i = \frac{2w_i(g_i - 1) - a_{in} - a_{ii} + a'_{ii} + 2w'_i}{2w'_i} = \frac{w_i}{w'_i}(g_i - 1) + 1 + \frac{a_{in}^2 - w_n a_{in}}{2w'_i w_n}$$

It is elementary to check that this is an integer ≥ 0 if we choose w'_i as in (4). □

Lemma 1.10. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type. Let e be the number of pairs (i,j) with $i < j$ and $a_{ij} > 0$. Then the expression $g_{top} = 1 - n + e$ is ≥ 0.*

Proof. If not, then $e < n-1$ which means there exists an i such that $a_{ij} = 0$ for all $j \neq i$. This contradicts assumption (3) of Definition 1.1. □

Definition 1.11. Let n, m_i, a_{ij}, w_i, g_i be a numerical type T. The *topological genus of T* is the nonnegative integer $g_{top} = 1 - n + e$ from Lemma 1.10.

We want to bound the genus by the topological genus. However, this will not always be the case, for example for numerical types with $n = 1$ as in Lemma 1.5. But it will be true for minimal numerical types which are defined as follows.

Definition 1.12. We say the numerical type n, m_i, a_{ij}, w_i, g_i of genus g is *minimal* if there does not exist an i with $g_i = 0$ and $a_{ii} = -w_i$, in other words, if there does not exist a (-1)-index.

We will prove that the genus g of a minimal type with $n > 1$ is greater than or equal to $\max(1, g_{top})$.

Lemma 1.13. *If n, m_i, a_{ij}, w_i, g_i is a minimal numerical type with $n > 1$, then $g \geq 1$.*

Proof. This is true because $g = 1 + \sum \Phi_i$ with $\Phi_i = m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ nonnegative by Lemma 1.7 and the definition of minimal types. □

Lemma 1.14 (Genus zero)**.** *The only minimal numerical type of genus zero is $n = 1$, $m_1 = 1$, $a_{11} = 0$, $w_1 = 1$, $g_1 = 0$.*

Proof. Follows from Lemmas 1.13 and 1.5. □

2. Models

In this chapter R will be a discrete valuation ring and K will be its fraction field. If needed we will denote $\pi \in R$ a uniformizer and $k = R/(\pi)$ its residue field.

Let V be an algebraic K-scheme (Varieties, Definition 06LG). A *model* for V will mean a flat finite type[1] morphism $X \to \mathrm{Spec}(R)$ endowed with an isomorphism $V \to X_K = X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(K)$. We often will identify V and the generic fibre X_K of X and just write $V = X_K$. The special fibre is $X_k = X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$. A *morphism of models* $X \to X'$ *for* V is a morphism $X \to X'$ of schemes over R which induces the identity on V.

We will say X *is a proper model of* V if X is a model of V and the structure morphism $X \to \mathrm{Spec}(R)$ is proper. Similarly for separated models, smooth models, and add more here. We will say X *is a regular model of* V if X is a model of V and X is a regular scheme. Similarly for normal models, reduced models, and add more here.

Definition 2.1. Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$. A *minimal model* will be a regular, proper model X for C such that X does not contain an exceptional curve of the first kind (Resolution of Surfaces, Section 0C2I).

Situation 2.2. Let R be a discrete valuation ring with fraction field K, residue field k, and uniformizer π. Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$. Let X be a regular proper model of C. Let $C_1, \ldots, C_n$ be the irreducible components of the special fibre X_k. Write $X_k = \sum m_i C_i$ as in Lemma 0C5Z.

Lemma 2.3. *In Situation 2.2 with* $\kappa_i = H^0(C_i, \mathcal{O}_{C_i})$ *and* g_i *the genus of* C_i *the data*

$$n, m_i, (C_i \cdot C_j), [\kappa_i : k], g_i$$

is a numerical type of genus equal to the genus of C.

Proof. (In the proof of Lemma 0CA3 we have seen that the quantities used in the statement of the lemma are well defined.) We have to verify the conditions (1) – (5) of Definition 1.1.

Condition (1) is immediate.

Condition (2). Symmetry of the matrix $(C_i \cdot C_j)$ follows from Equation (0C65) and Lemma 0C64. Nonnegativity of $(C_i \cdot C_j)$ for $i \neq j$ is part (1) of Lemma 0C66.

Condition (3) is part (3) of Lemma 0C66.

Condition (4) is part (2) of Lemma 0C66.

Condition (5) follows from the fact that $(C_i \cdot C_j)$ is the degree of an invertible module on C_i which is divisible by $[\kappa_i : k]$, see Varieties, Lemma 0C6P.

The genus formula proved in Lemma 0CA3 tells us that the numerical type has the genus as stated, see Definition 1.4. □

Definition 2.4. In Situation 2.2 the *numerical type associated to* X is the numerical type described in Lemma 2.3.

Now we match minimality of the model with minimality of the type.

Lemma 2.5. *In Situation 2.2. The following are equivalent*

(1) X *is a minimal model, and*
(2) *the numerical type associated to* X *is minimal.*

[1] Occasionally it is useful to allow models to be locally of finite type over R, but we'll cross that bridge when we come to it.

Proof. If the numerical type is minimal, then there is no i with $g_i = 0$ and $(C_i \cdot C_i) = -[\kappa_i : k]$, see Definition 1.12. Certainly, this implies that none of the curves C_i are exceptional curves of the first kind.

Conversely, suppose that the numerical type is not minimal. Then there exists an i such that $g_i = 0$ and $(C_i \cdot C_i) = -[\kappa_i : k]$. We claim this implies that C_i is an exceptional curve of the first kind. Namely, the invertible sheaf $\mathcal{O}_X(-C_i)|_{C_i}$ has degree $-(C_i \cdot C_i) = [\kappa_i : k]$ when C_i is viewed as a proper curve over k, hence has degree 1 when C_i is viewed as a proper curve over κ_i. Applying Algebraic Curves, Proposition 0C6U we conclude that $C_i \cong \mathbf{P}^1_{\kappa_i}$ as schemes over κ_i. Since the Picard group of $\mathbf{P}^1$ over a field is $\mathbf{Z}$, we see that the normal sheaf of C_i in X is isomorphic to $\mathcal{O}_{\mathbf{P}_{\kappa_i}}(-1)$ and the proof is complete. □

Lemma 2.6. *Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$ and genus 0. If there is more than one minimal model for C, then the special fibre of every minimal model is isomorphic to $\mathbf{P}^1_k$.*

Proof. Let X be some minimal model of C. The numerical type associated to X has genus 0 and is minimal (Definition 2.4 and Lemma 2.5). Hence by Lemma 1.14 we see that X_k is reduced, irreducible, has $H^0(X_k, \mathcal{O}_{X_k}) = k$, and has genus 0. Let Y be a second minimal model for C which is not isomorphic to X. By Resolution of Surfaces, Lemma 0C5S there exists a diagram of S-morphisms

$$X = X_0 \leftarrow X_1 \leftarrow \ldots \leftarrow X_n = Y_m \rightarrow \ldots \rightarrow Y_1 \rightarrow Y_0 = Y$$

where each morphism is a blowup in a closed point. We will prove the lemma by induction on m. The base case is $m = 0$; it is true in this case because we assumed that Y is minimal hence this would mean $n = 0$, but X is not isomorphic to Y, so this does not happen, i.e., there is nothing to check.

Before we continue, note that $n + 1 = m + 1$ is equal to the number of irreducible components of the special fibre of $X_n = Y_m$ because both X_k and Y_k are irreducible. Another observation we will use below is that if $X' \rightarrow X''$ is a morphism of regular proper models for C, then $X' \rightarrow X''$ is an isomorphism over an open set of X'' whose complement is a finite set of closed points of the special fibre of X'', see Varieties, Lemma 0BFP. In fact, any such $X' \rightarrow X''$ is a sequence of blowing ups in closed points (Resolution of Surfaces, Lemma 0C5R) and the number of blowups is the difference in the number of irreducible components of the special fibres of X' and X''.

Let $E_i \subset Y_i$, $m \geq i \geq 1$ be the curve which is contracted by the morphism $Y_i \rightarrow Y_{i-1}$. Let i be the biggest index such that E_i has multiplicity > 1 in the special fibre of Y_i. Then the further blowups $Y_m \rightarrow \ldots \rightarrow Y_{i+1} \rightarrow Y_i$ are isomorphisms over E_i since otherwise E_j for some $j > i$ would have multiplicity > 1. Let $E \subset Y_m$ be the inverse image of E_i. By what we just said $E \subset Y_m$ is an exceptional curve of the first kind. Let $Y_m \rightarrow Y'$ be the contraction of E (which exists by Resolution of Surfaces, Lemma 0C2N). The morphism $Y_m \rightarrow X$ has to contract E, because X_k is reduced. Hence there are morphisms $Y' \rightarrow Y$ and $Y' \rightarrow X$ (by Resolution of Surfaces, Lemma 0C5J) which are compositions of at most $n - 1 = m - 1$ contractions of exceptional curves (see discussion above). We win by induction on m. Upshot: we may assume that the special fibres of all of the curves X_i and Y_i are reduced.

Since the fibres of X_i and Y_i are reduced, it has to be the case that the blowups $X_i \rightarrow X_{i-1}$ and $Y_i \rightarrow Y_{i-1}$ happen in closed points which are regular points of the special fibres. Namely, if X'' is a regular model for C and if $x \in X''$ is a closed point of the special fibre, and $\pi \in \mathfrak{m}_x^2$, then the exceptional fibre E of the blowup

$X' \to X''$ at x has multiplicity at least 2 in the special fibre of X' (local computation omitted). Hence $\mathcal{O}_{X''_k,x} = \mathcal{O}_{X'',x}/\pi$ is regular (Algebra, Lemma 00NQ) as claimed. In particular x is a Cartier divisor on the unique irreducible component Z' of X''_k it lies on (Varieties, Lemma 0B8Y). It follows that the strict transform $Z \subset X'$ of Z' maps isomorphically to Z' (use Divisors, Lemmas 080E and 0807). In other words, if an irreducible component Z of X_i is not contracted under the map $X_i \to X_j$ $(i > j)$ then it maps isomorphically to its image.

Now we are ready to prove the lemma. Let $E \subset Y_m$ be the exceptional curve of the first kind which is contracted by the morphism $Y_m \to Y_{m-1}$. If E is contracted by the morphism $Y_m = X_n \to X$, then there is a factorization $Y_{m-1} \to X$ (Resolution of Surfaces, Lemma 0C5J) and moreover $Y_{m-1} \to X$ is a sequence of blowups in closed points (Resolution of Surfaces, Lemma 0C5R). In this case we lower m and we win by induction. Finally, assume that E is not contracted by the morphism $Y_m \to X$. Then $E \to X_k$ is surjective as X_k is irreducible and by the above this means it is an isomorphism. Hence X_k is isomorphic to a projective line as desired. $\square$